Imitation Learning

Kianté Brantley
CS 1840 Fall 2026

Announcements

HW3 was due last Friday!

HW4 is out!

HW5 is a mini-project in groups of 2–4.
For HW4, submit your group members and proposed project ideas.

Check the course website for Office Hours:
https://harvard-cs-1840.github.io/2026/

Recap

Infinite horizon Discounted MDPs

$$\mathscr{M} = \{S, A, \gamma, r, P, \mu\}$$

Average state distribution: $d^{\pi} = (1-\gamma)\sum_{h=0}^{\infty} \gamma^h \mathbb{P}_h^{\pi}$

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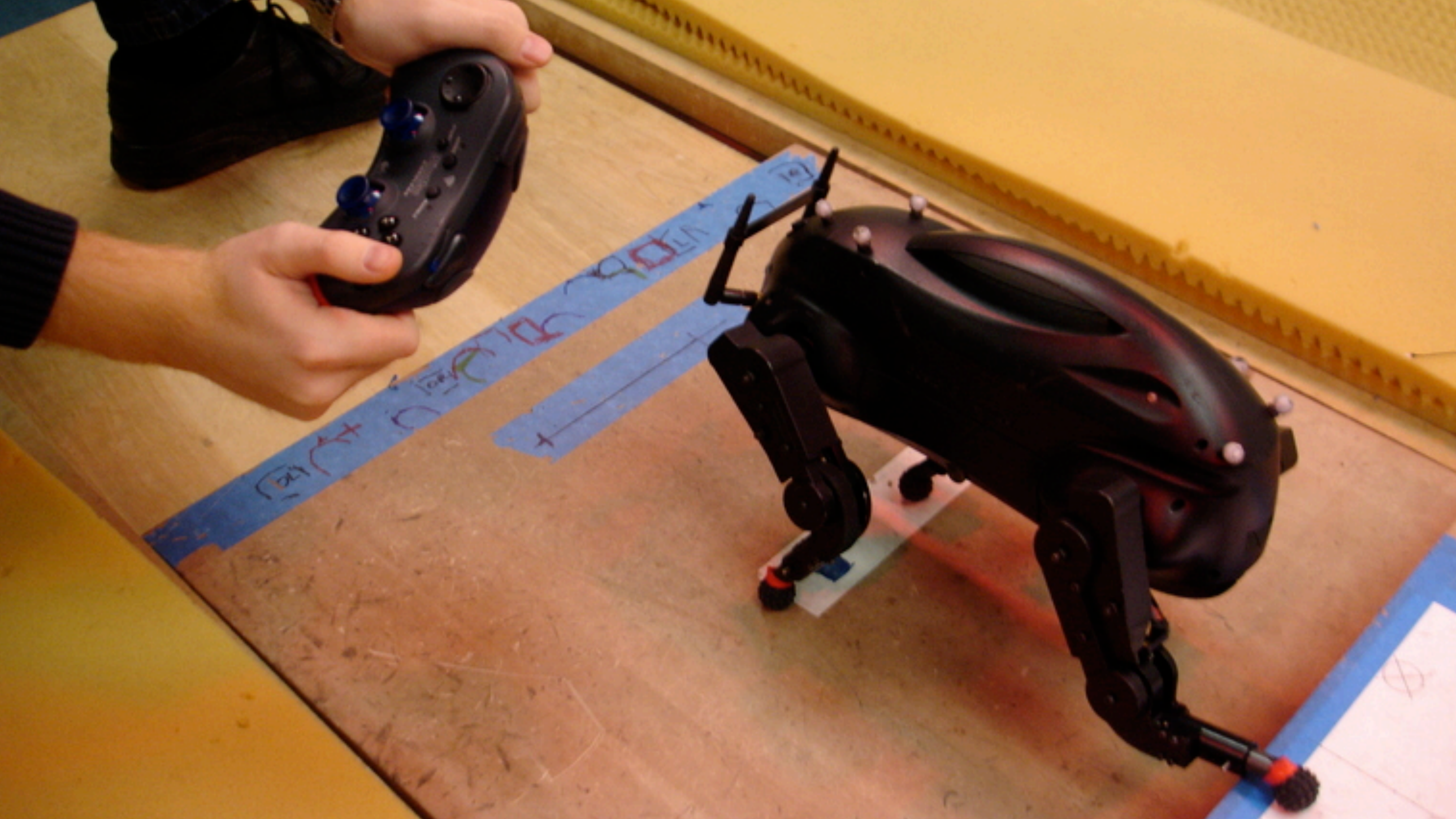
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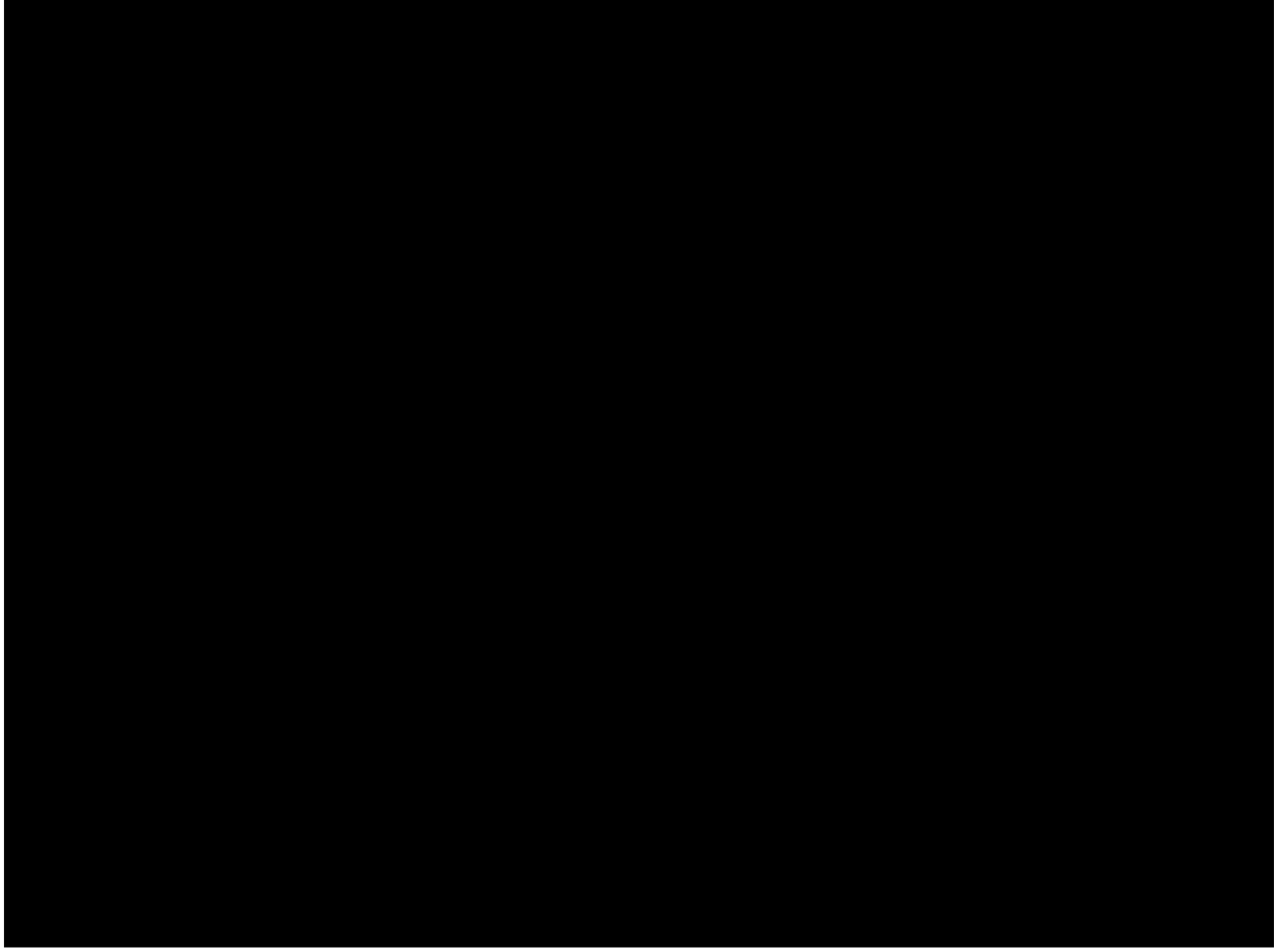
Advantage: $A^\pi := Q^\pi(s,a) - V^\pi(s,a)$

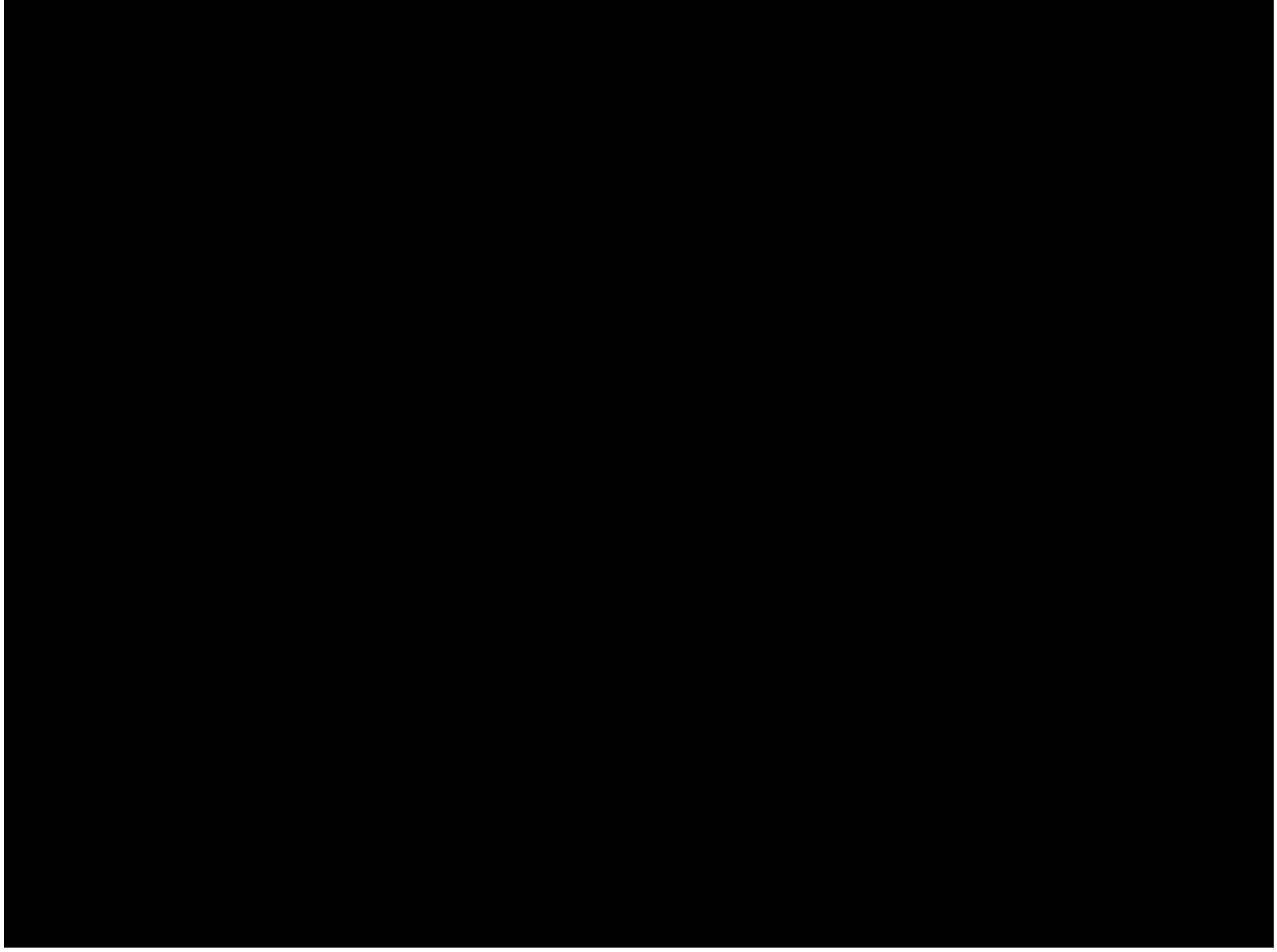
Outline:

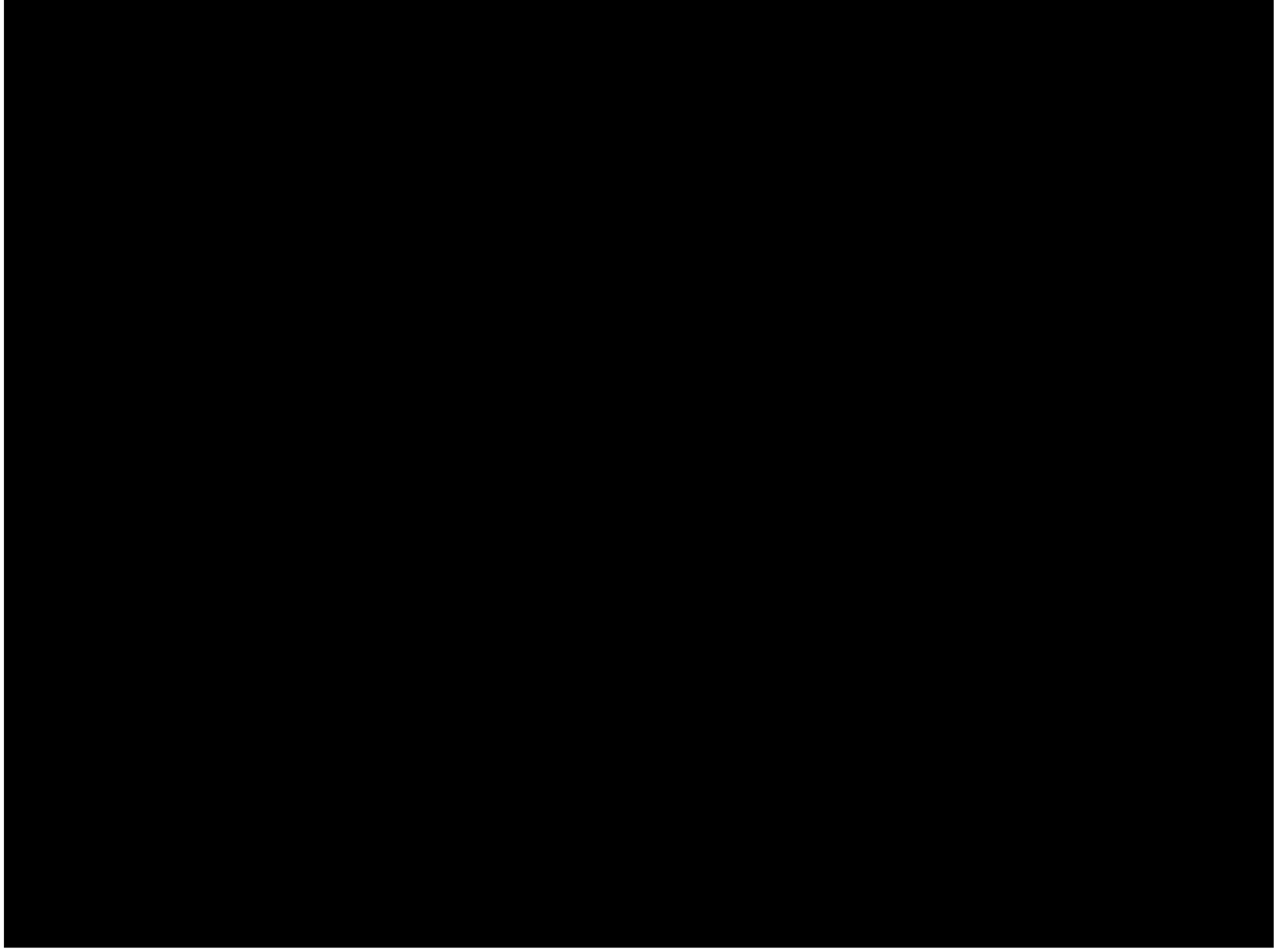
1. Offline Imitation Learning: Behavior Cloning

2. Performance difference lemma and its application to proving BC’s bound









An Autonomous Land Vehicle

In A Neural Network [Pomerleau, NIPS ‘88]



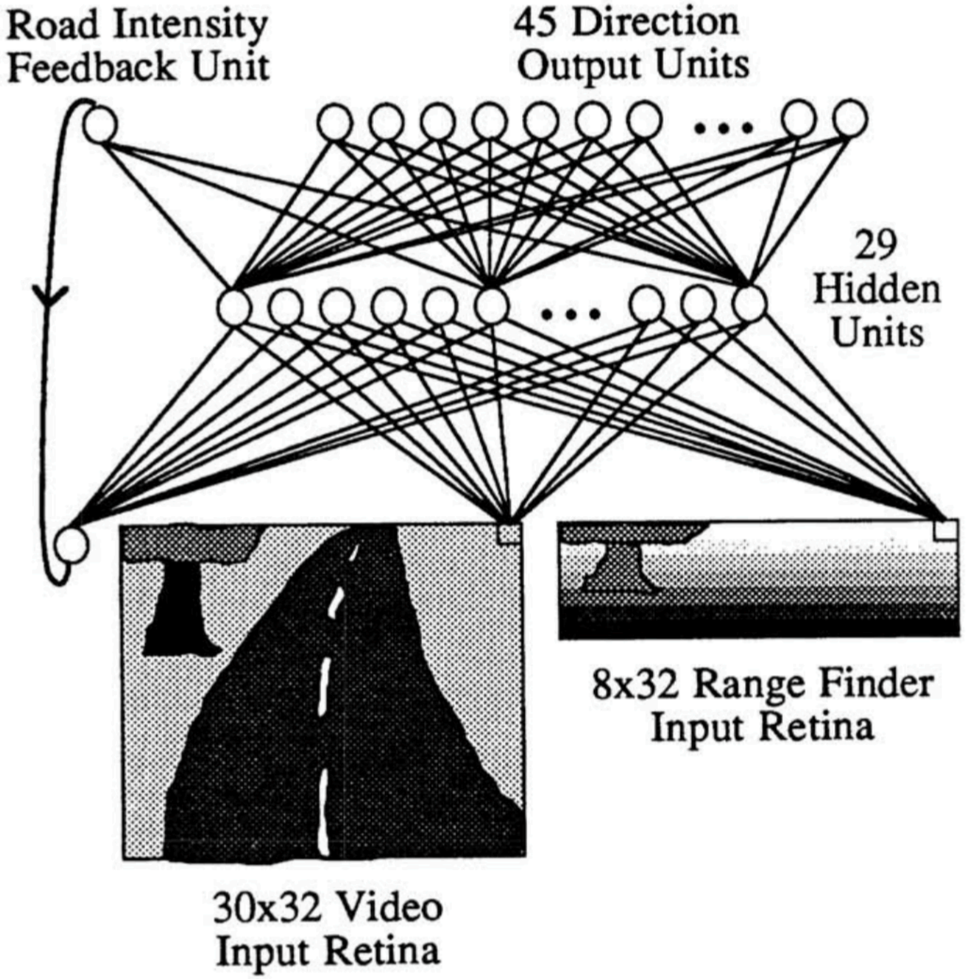
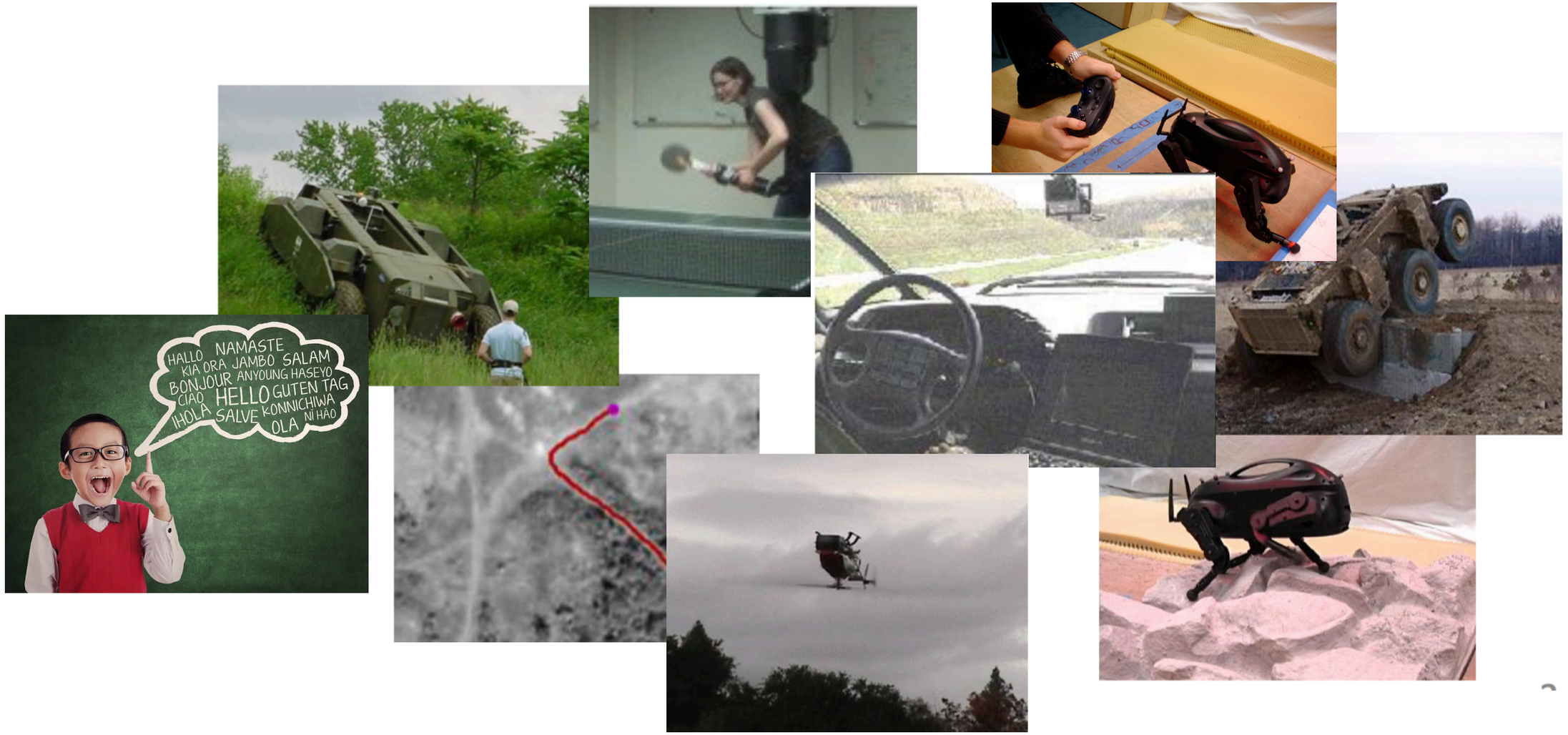


Figure 1: ALVINN Architecture

Imitation Learning



Imitation Learning



Imitation Learning

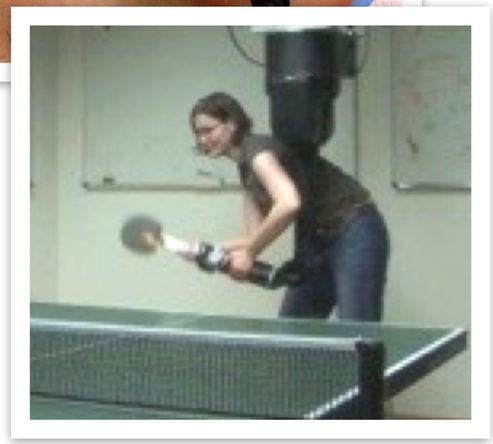
Imitation Learning



Imitation Learning

Expert Demonstrations



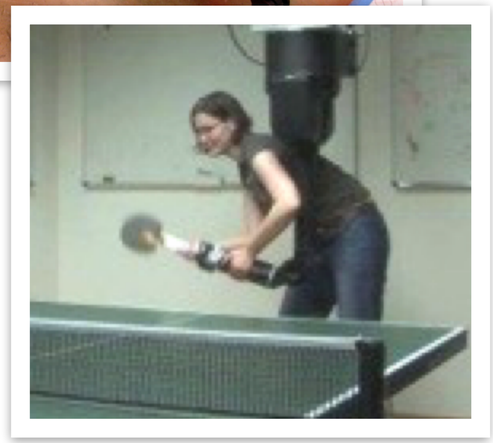


Imitation Learning

Expert Demonstrations

Machine Learning Algorithm



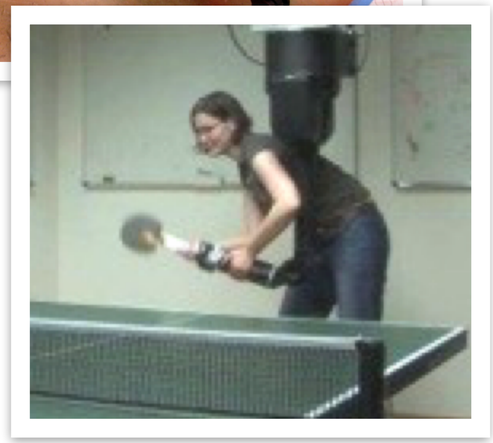


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- Gaussian Process
- Kernel Estimator
- Deep Networks
- Random Forests
- LWR
- …

Imitation Learning

Expert Demonstrations





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Policy π

Maps *states* to actions

Learning to Drive by Imitation

[Pomerleau89, Saxena05, Ross11a]

Input:



Camera Image

Policy

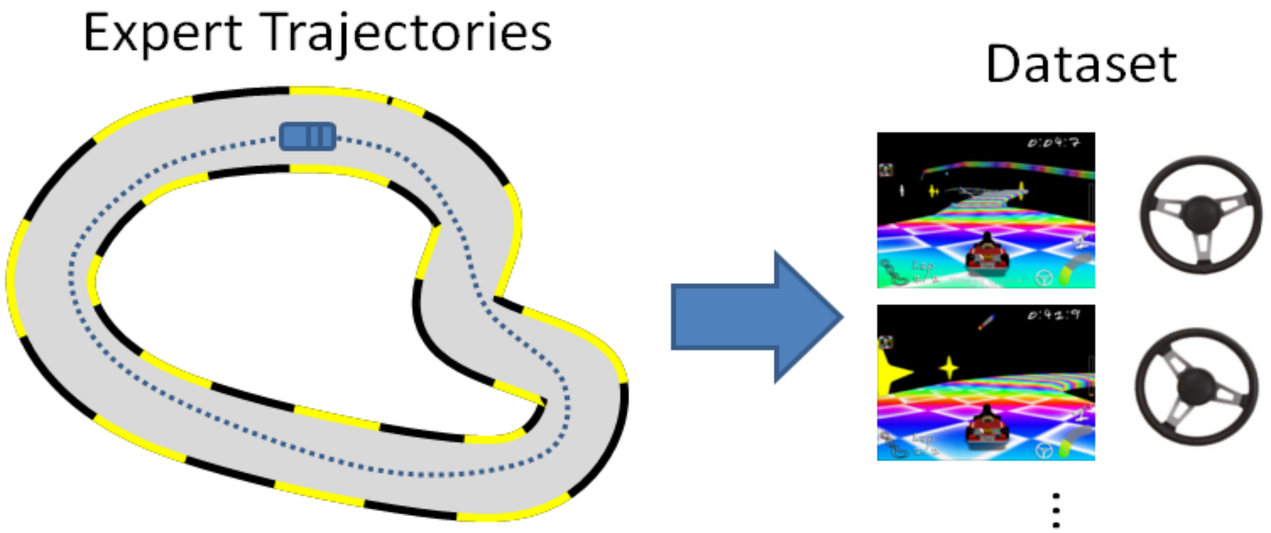
Output:



Steering Angle
in [-1, 1]

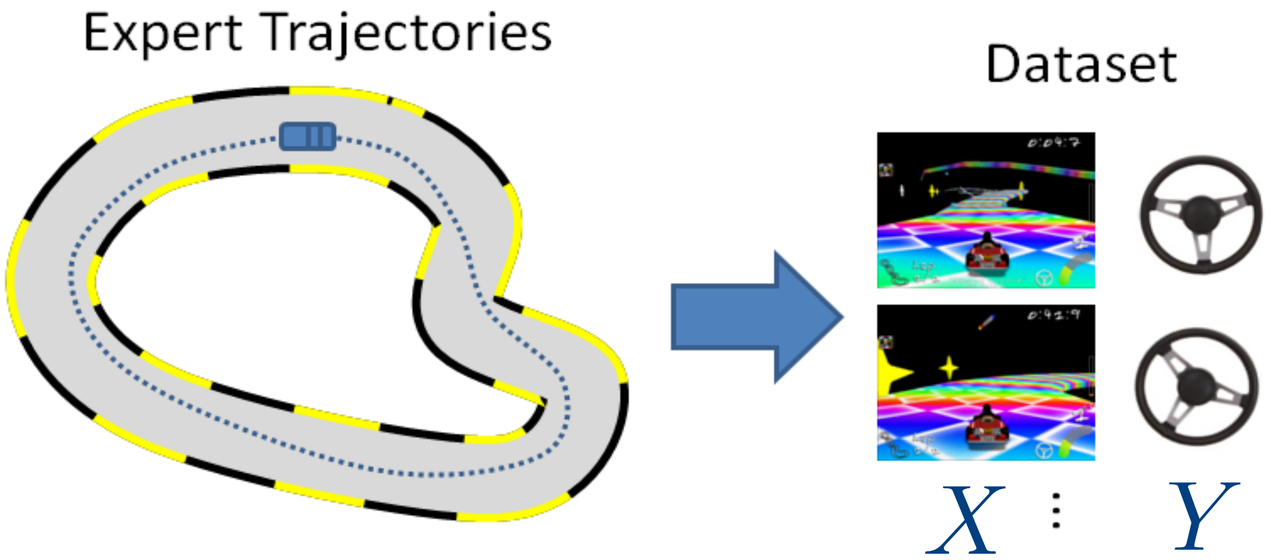
[Widrow64,Pomerleau89]

Supervised Learning Approach: Behavior Cloning



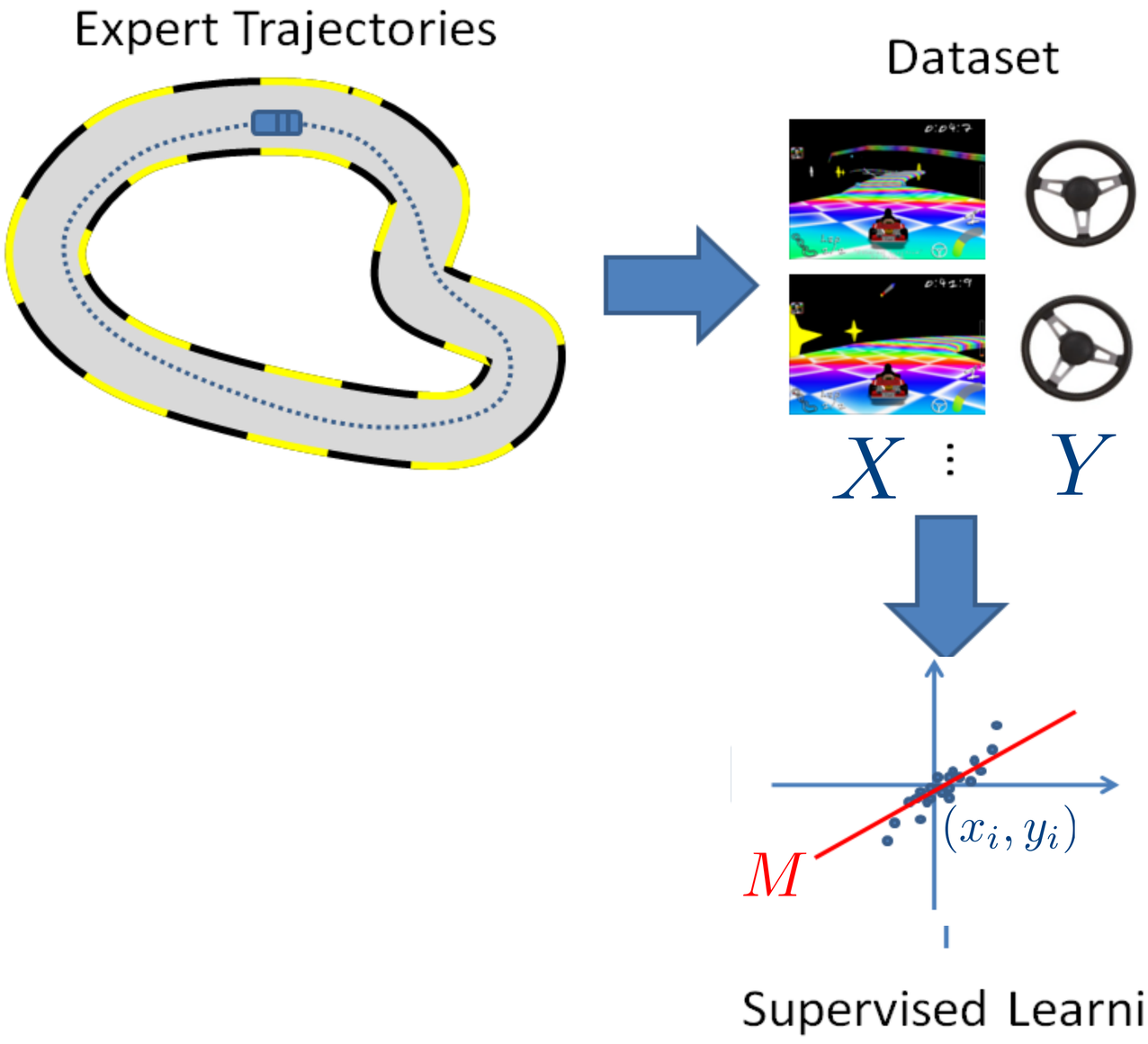
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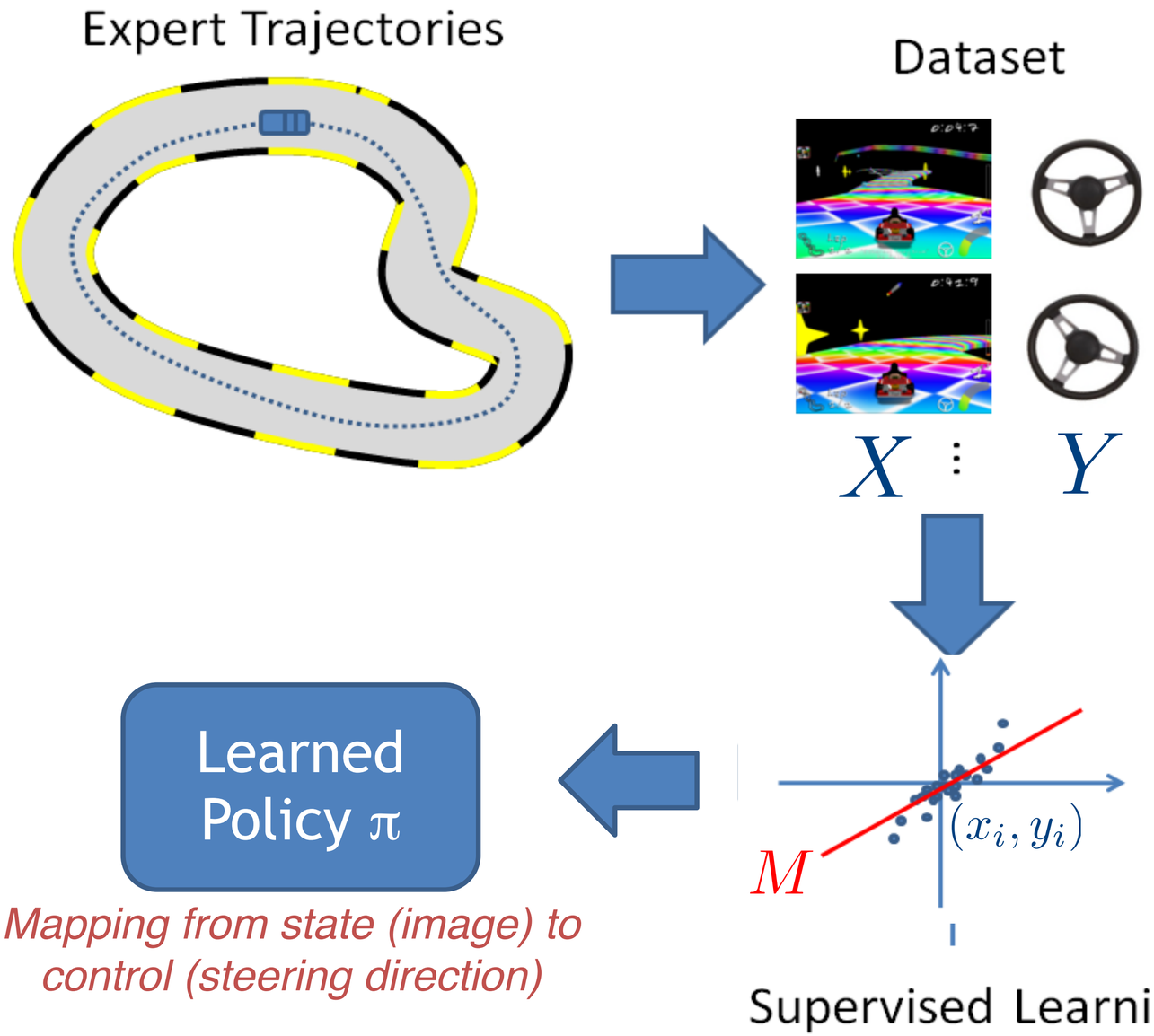
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Ready!


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LLMs are trained via BC in their pre-training phase

Take a sentence from the web:

State Action

Reinforcement learning (RL) is an interdisciplinary area of machine learning and optimal control concerned with how an intelligent agent should take actions in a dynamic environment in order to maximize a reward signal.

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Reinforcement learning (RL) is an interdisciplinary area of machine learning and optimal control concerned with how an intelligent agent should take actions in a dynamic environment in order to maximize a reward signal.

Forcing LLM to predict the next “action” conditioned on past…

Let’s formalize the offline IL Setting

and the Behavior Cloning algorithm

Discounted infinite horizon MDP $\mathscr{M} = \{S, A, \gamma, r, P, \rho, \pi^\star\}$

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Goal: learn a policy from $\mathscr{D}$ that is as good as the expert $\pi^\star$

Let's formalize the Behavior Cloning algorithm

BC Algorithm input: a restricted policy class $\Pi = \{\pi : S \mapsto \Delta(A)\}$

BC is a Reduction to Supervised Learning:

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2. square loss (i.e., regression for continuous action): $\ell(\pi, s^\star, a^\star) = \|\pi(s^\star) - a^\star\|_2^2$

Analysis

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Does that imply that $\hat{\pi}$ is a good policy? What's the performance difference between $\hat{\pi}$ and $\pi^\star$?

Outline:



1. Offline Imitation Learning: Behavior Cloning

2. Performance difference lemma and its application to proving BC’s bound

Performance Difference Lemma

Given two policies $\pi : S \mapsto \Delta(A),\ \pi' : S \mapsto \Delta(A)$, recall $V^{\pi}(s_0) = \mathbb{E}\left[\sum_{h=0}^{\infty} \gamma^h r(s_h, a_h) \,|\, \pi\right]$

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Performance Difference Lemma (PDL):

$$V^{\pi}(s_0) - V^{\pi'}(s_0) = \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi}_{s_0}}\left[\mathbb{E}_{a\sim\pi(\cdot|s)}Q^{\pi'}(s,a) - V^{\pi'}(s)\right]$$

$$:= \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi}_{s_0}}\left[\mathbb{E}_{a\sim\pi(\cdot|s)}A^{\pi'}(s,a)\right]$$

PDL Explanation

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Proof of Sketch (see reading material for detailed steps)

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$$(1-\gamma)\left(V^\star - V^{\widehat{\pi}}\right) = \mathbb{E}_{s\sim d^{\pi^\star}} A^{\widehat{\pi}}(s, \pi^\star(s))$$

$$= \mathbb{E}_{s\sim d^{\pi^\star}} A^{\widehat{\pi}}(s, \pi^\star(s)) - \mathbb{E}_{s\sim d^{\pi^\star}} A^{\widehat{\pi}}(s, \widehat{\pi}(s))$$

$$\leq \mathbb{E}_{s\sim d^{\pi^\star}} \frac{2}{1-\gamma} \mathbf{1}\left\{\widehat{\pi}(s) \neq \pi^\star(s)\right\}$$

$$\leq \frac{2}{1-\gamma}\epsilon$$

The quadratic amplification is annoying;
Related to pre-trained LLM hallucination;
Will see how to fix it next lecture

Summary

BC: simple algorithm that directly learns from human demonstrations; used in robotics and NLP

PDL: how to capture the performance difference between two policies
(as important as Simulation lemma)

Feedback and Attendance

https://bit.ly/4gYn2nC

